

Unc-SOD: An Uncertainty Learning Framework for Small Object Detection

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Abstract—Small object detection (SOD) constitutes a notable yet immensely arduous task, stemming from the restricted informative regions inherent in size-limited instances, which further sparks off heightened uncertainty beyond the capacity of current two-stage detectors. Specifically, the intrinsic ambiguity in small objects undermines the prevailing sampling paradigms and may mislead the model to devote futile effort to those unrecognizable targets, while the inconsistency of features utilized for the detection at two stages further exposes the hierarchical uncertainty. In this paper, we develop an Uncertainty learning framework for Small Object Detection, dubbed as Unc-SOD. By incorporating an auxiliary uncertainty branch to conventional Region Proposal Network (RPN), we model the indeterminacy at instance-level which later on serves as a surrogate criterion for sampling, thereby unearthing adequate candidates dynamically based on the varying degrees of uncertainty and facilitating the learning of proposal networks. In parallel, a Perception-and-Interaction strategy is devised to capture rich and discriminative representations, through optimizing the intrinsic properties from the regional features at the original pyramid and the assigned one, in which the perceptual process unfolds in a mutual paradigm. As the seminal attempt to model uncertainty in SOD task, our Unc-SOD yields state-of-the-art performance on two large-scale small object detection benchmarks, SODA-D and SODA-A, and the results on several SOD-oriented datasets including COCO, VisDrone, and Tsinghua-Tencent 100K also exhibit the promotion to baseline detector. This underscores the efficacy of our approach and its superiority over prevailing detectors when dealing with small instances.

Index Terms—Small object detection, uncertainty learning, sampling strategy, feature interaction.

I. INTRODUCTION

Small object detection (SOD) is a sub-task of generic object detection, which aims to perceive the size-limited instances and serves a pivotal function in autonomous driving, medical image analysis, vision-based surveillance system, to name a few [1], [2], [3], [4], [5]. In comparison to generic object detection that is recently dominated by DETR [6] and its variants [7], [8], [9], two-stage

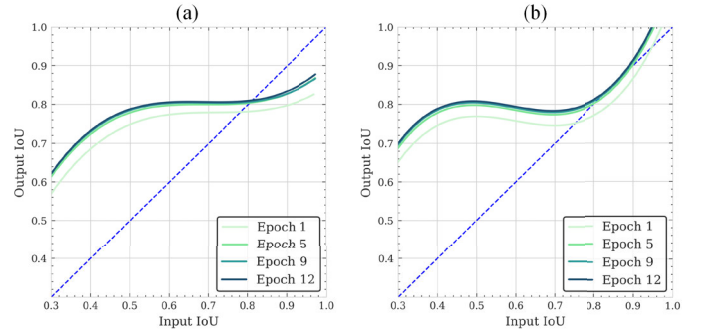


Fig. 1. The regression quality of RPN with Maximum-IoU strategy (a) and with our method (b) for *Small* instances in the SODA-D dataset [2] along the training goes. It can be observed that, for both of these strategies, the localization quality of positive priors hardly improves after just a few epochs.

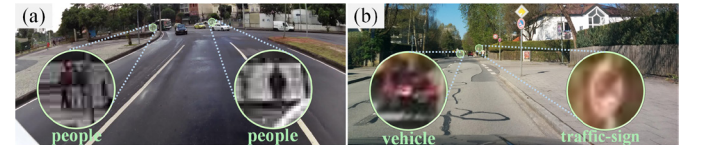


Fig. 2. The structural distortion of small instances induces inherent ambiguity to the annotation and regression. One can hardly capture accurate bounding boxes of the *people* in (a), and the identification of the objects *vehicle* and *traffic-sign* in (b) poses severe challenges to the detector, as the essential discriminative cues are rendered non-existent. Best viewed in zoom-in windows.

convolution neural networks (CNN)-based paradigms [10], [11] showcase a considerable edge over other competitors [6], [12], [13], [14], [15] in detecting small objects [16]. However, the uncertainty inherent in small instances remains a bottleneck in the optimization of proposal-based detectors.

Region proposal network (RPN) is designed for foreground-background classification and regression on top of the dense priors. Albeit such prior-to-proposal pipeline has proven to be effective in generic detection tasks [10], [11], [17], it encounters unexpected hurdles during generating proposals for small instances. As depicted in Fig. 1, the regression performance begins to saturate as the training progresses into the 5-th epoch, and such saturation can be attributed to the inherent **uncertainty at data-level**. To be specific, small objects usually encapsulate a distorted structure due to partial occlusion, motion blur and lens flare, *etc* [18], [19]. Hence, delineating the boundaries of these size-limited instances and annotating them accurately present a daunting task [2], [20], [21], see *people* in Fig. 2 (a). Essentially, such ambiguity is ubiquitous in the mainstream datasets [22] and inhibits the

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optimization of the localization network, because pioneering detectors [10], [12], [17] fail to consider the uncertainty of ground truth (GT) objects. Observing this, previous works [23], [24], [25] handle the regression variance by formulating the boxes as Gaussian or general distribution. Albeit granting access to uncertainty modeling of box locations and achieving reliable predictions, these approaches fall short in handling the objects with severe deformation or extremely sizes which may become unrecognizable, as shown in Fig. 2 (b). Worse still, while one would expect nuanced treatment based on varying degrees of uncertainty for different objects, the prevailing proposal paradigms [10], [26], [27], instead, resort to a one-size-fits-all strategy. Concretely, they employ the rigid Intersection-over-Union (IoU) criterion (say $\text{IoU} \geq 0.7$) to determine the positive/negative attributes of priors, and among them only scarce number of anchors are registered as positive ones. Such paucity of samples further restrains the optimization of proposal networks [16], [28], and sacrifices the regression of high-quality priors as exhibited in Fig. 1 (a), where the priors with an input IoU larger than 0.8 tend to move even farther away from the target instead of getting closer, which is paradoxical to the intention of RPN.

After obtaining proposals, two-stage detectors [10], [11], [17] develop a heuristic and size-dependent mapping strategy to specify the corresponding pyramidal levels, on which RoI Align is exploited to achieve regional features for subsequent detection. By this setup, the objects at various scales are allocated to appropriate layers, facilitating the explicit and simultaneous optimization of hierarchical pyramids [2], [18]. A common rationale behind this divide-and-conquer paradigm lies in that high-level features offer a broader receptive field and more compact representations, proving beneficial for large-scale objects, while the detection of smaller instances entails edge, texture, and other fine-grained responses, which are often embraced by shallow layers [17], [18]. But the situation worsens for SOD task, where regional representations are derived solely from the lowest pyramidal level (generally P_2), leading the model to update features of other levels through an implicit and indirect fashion [2], [29]. In parallel, the proposals are evolved from the priors at multiple pyramids, which implies that the hierarchical features employed for classification and regression across two stages often exhibit incongruities. We map the level distribution of proposals emanated from Faster RCNN [10] in Fig. 3, with over 95% originating from higher levels, namely $\{P_3, P_4, P_5\}$. Such inconsistency essentially brings to light a thorny yet underexplored issue in small object detection—the **uncertainty at hierarchy-level**. To elaborate further: which layer should we register to derive powerful representations for small instances?

Faced with the aforementioned barriers, we propose an **Uncertainty learning framework for Small Object Detection**, namely **Unc-SOD**. In order to instill the model with ambiguity awareness and alleviate futile investigation into unrecognizable samples, we extend the conventional RPN with an auxiliary prediction branch to model instance-level location uncertainty which then serves as a surrogate criterion to procure potential candidates and the routine is shown in Fig. 4, thereby mitigating the sample scarcity and facilitating the proposal generation quality (see Fig. 1 (b)), while preventing the optimization from being contaminated by the low-quality priors of highly

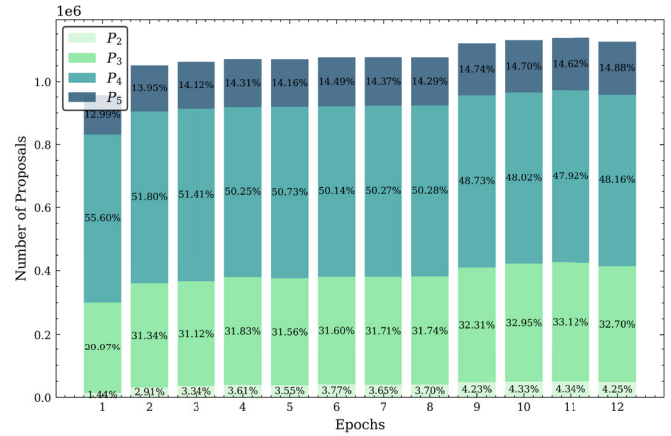


Fig. 3. Distribution of the pyramidal levels to proposals of faster RCNN [10] on the SODA-D dataset, and only about 5% of them originate from P_2 .

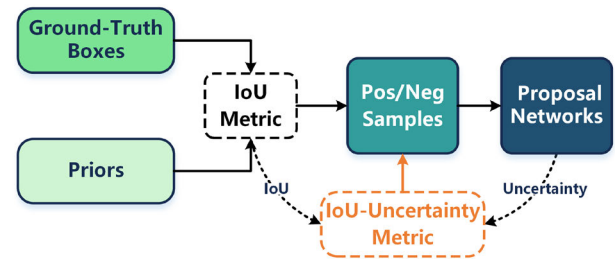


Fig. 4. Comparison between conventional sampling strategy and our uncertainty-aware solution, which is highlighted in the orange dashed box.

equivocal instances. When it comes to the hierarchy uncertainty, we devise a Perception-and-Interaction strategy to capture rich and discriminative representations by exploiting the intrinsic correlations between the regional features at the original level and the assigned one. The perceptual process unfolds in a reciprocal fashion, with analytic perception reinstating the structural details of high-level regional features, whereas holistic perception injects succinct patterns into the region representations from lower levels. Following that, we develop a cross-interaction scheme to re-calibrate the features at two levels and ultimately, acquire discriminative representations for small instances. The main contributions are as follows:

- Devising an Uncertainty-aware Sampling strategy to alleviate sample scarcity issue, where a novel branch is installed to RPN to model the uncertainty of GT instances, which will serve as a proxy criterion to procure more candidates. Our solution spawns sufficient proposals for size-limited objects while minimizing the risk of inherently unrecognizable instances misleading the model.
- Formulating a Perception-and-Interaction scheme to obtain discriminative representations for small instances via exploiting the intrinsic correlations between the region features from the original level and the assigned one.
- Our uncertainty-modeling framework for small object detection achieves state-of-the-art performance on the large-scale benchmark SODA. And the results on several SOD-oriented datasets including COCO, VisDrone, and Tsinghua-Tencent 100K underscore the generality and robustness of Unc-SOD across diverse scenes.

II. RELATED WORK

A. Small Object Detection

Detecting small objects is a notoriously intractable task due to their sparse visual cues and extremely limited informative regions. To address this challenge, researchers have explored several key approaches. Deng et al. [18] extended conventional FPN [17] with an extra high-resolution pyramid level tailored for detecting small instances. In a similar architectural redesign, HR-FPN [30] restructures the pyramid by upsampling all feature levels to a single, unified high resolution, better serving SOD task in drone views. Drawing inspiration from human cognition, Kim et al. [19] uniquely devised memory networks to recall large-scale exemplars from small-scale cues. SR-TOD [31] leverages the pixel-level difference between the original image and its reconstructed counterpart to boost the representations. Bian et al. [32] leveraged information theory to identify and boost weak regions via entropy modeling. Concentrating on the frequency domain, Sun et al. [33] enhanced spectral signatures through background smoothing and adversarial perturbations. Beyond feature-level refinement, another paradigm modifies the inference architecture itself for efficiency. AutoFocus [34] designs a coarse-to-fine scheme to selectively process high-resolution chips, while Hong et al. [35] employed a reinforcement learning agent to dynamically control a multi-branch detection framework.

Due to the extremely limited resolutions, the pioneering sampling paradigms [36], [37], [38] in generic object detectors fail to assign adequate samples for small instances, rendering sample scarcity a primary bottleneck. To this end, Kisantal et al. utilize copy-and-paste techniques, while DS-GAN [39] leverages data synthesis for augmentation. Regarding assignment strategies, Zhu et al. [40] developed Expected Max Overlapping to calculate the overlaps between ground truths and priors. RFLA [28] measures the similarity between the Gaussian receptive field and GT boxes in the assigning process. Following this line, SimD [41] proposes to evaluate location and shape similarity for better assignment. Dai et al. [3] introduced Adaptive Sample Assignment Strategy (ASAS) to focus valuable foreground areas based on the predicted heatmaps. CFNet [16] employs a coarse-to-fine regression scheme, in which a dynamic criterion is introduced to mine potential candidates in conventional prior-to-proposal pipelines.

Although the aforementioned solutions partially mitigate the sample paucity problem, they fail to respond to the uncertainty inherent in small objects, which has a detrimental effect on the learning process. In contrast, our strategy can dynamically exploit these ambiguities for optimal assignment and unearth potential candidates for training.

B. Uncertainty-Modeling in Object Detection

Datasets inherently hold uncertainty and ambiguity, attributable to the diverse conditions under which images are captured and the fluctuations in annotations. In addition, safety-critical scenarios such as autonomous driving and surveillance necessarily entail reliable localization predictions to guarantee safe and optimal decision-making [24], [42]. To this end, He et al. [23] formulated the bounding boxes and

GT boxes as Gaussian distributions and Dirac delta functions, respectively. Then the regression could be accomplished by minimizing the Kullback-Leibler Divergence between them, and the predicted standard deviations will be utilized to improve Non-Maximum Suppression (NMS) by highlighting those boxes with lower uncertainties. Gaussian-YOLOv3 [24] models the box coordinates as Gaussian parameters to achieve an outstanding real-time detector which reduces False Positives by estimating the uncertainty during box regression. GFL [25] directly learns the discretized probability distribution instead of introducing stronger priors which may fail to capture the real distribution of the locations of bounding boxes. Deeming that previous uncertainty-modeling frameworks who train variance branch with negative log likelihood (NLL) may carry the risk of high entropy predictive distributions, Harakeh et al. [42] leveraged the energy score as a non-local proper scoring rule for training and proposed an alternate evaluation metric based on scoring rules during evaluating.

Moreover, uncertainty modeling has been explored for various detection sub-domains. The majority of these works focus on aleatoric uncertainty, which captures inherent data noise. For instance, Park et al. [43] innovatively proposed a cross-modal learning strategy by measuring the RoI uncertainty, thereby constructing a robust multispectral pedestrian detection framework. Liu et al. [44] handled noisy labels in camouflaged object detection by training an uncertainty network with an explicit supervision. For future vehicle localization, Hudnell et al. [45] derived the uncertainty from a robust Huber loss to serve as a confidence measure. In the anchor-free paradigm, Lee et al. [46] introduced the Negative Power Log-likelihood (NPLL) loss to learn the uncertainty, which then modulates the classification loss to align the classification score with localization quality. Beyond aleatoric uncertainty, epistemic uncertainty, which concentrates on the model's own predictive ambiguity [47], [48], has also been explored. A novel Class Uncertainty-Aware (CUA) loss [49] was proposed to modulate loss gradients directly with the model's epistemic uncertainty, compelling the detector to focus on its unreliable predictions. Demonstrating the synergy of both types, Kim et al. [50] developed a unified framework for multispectral pedestrian detection. Specifically, they leveraged aleatoric RoI uncertainty for robust feature fusion against camera miscalibration, while the predictive uncertainty was used to steer the feature distribution alignment between different modalities, effectively mitigating the cross-modal discrepancy issue.

Though these pioneering efforts investigate aleatoric uncertainty and epistemic uncertainty in detection paradigms, they falter when confronted with small objects [2] due to the extremely degraded situation. In comparison, our approach incorporates the uncertainty into sampling and gleans off-the-shelf cues for regional representations, leading to a powerful two-stage small object detector.

III. OUR APPROACH

This section illuminates the details of our proposed Unc-SOD which is exhibited in Fig. 5. We first present the localization deviation branch along with the corresponding uncertainty-aware sampling strategy. Next, the structural pipeline of Perception-and-Interaction is elucidated.

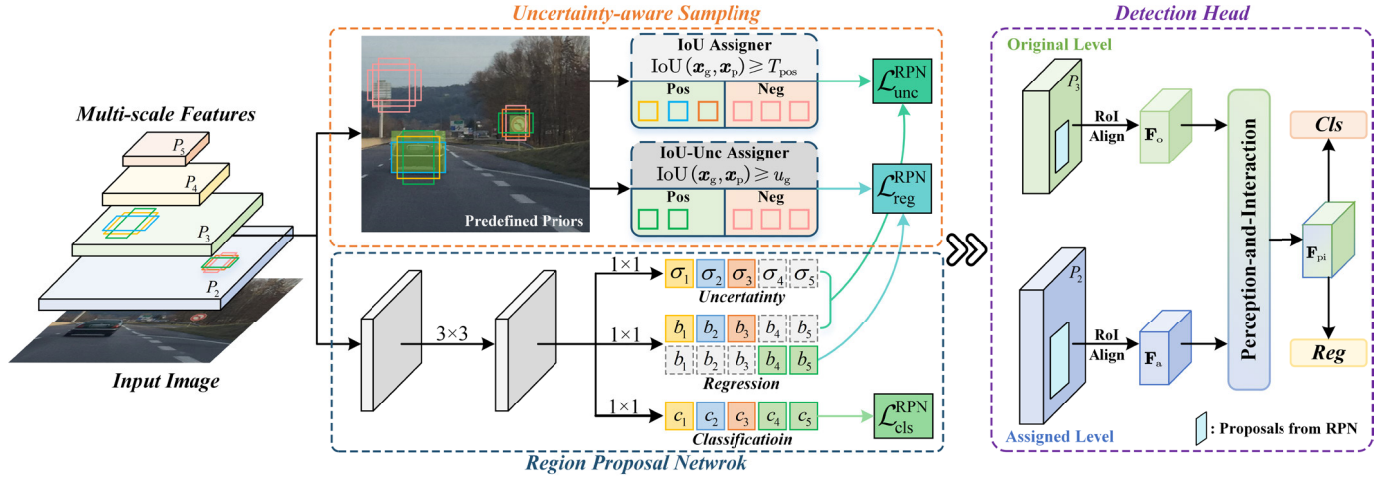


Fig. 5. Schematic pipeline of Unc-SOD, which contains two major components: the region proposal network and the detection head. For proposal generation part, we extend RPN with an auxiliary branch to predict the standard deviation of positive priors and further procure the instance-level uncertainty, which then serves as a surrogate criterion to unearth potential candidates. $\mathcal{L}_{\text{cls}}^{\text{RPN}}$, $\mathcal{L}_{\text{reg}}^{\text{RPN}}$, and $\mathcal{L}_{\text{unc}}^{\text{RPN}}$ intend to optimize the classification, regression, and uncertainty branch in RPN, respectively. After that, for each proposal emanated from RPN, we extract region representations from both the pyramidal level assigned by the heuristic size-dependent mapping strategy (F_a) and the corresponding level associated with the original anchor of the proposal (F_o). Then, these two counterpart features are fed to the proposed Perception-and-Interaction module simultaneously, which ultimately derives discriminative representations of small instances, denoted as F_{pi} , for subsequent classification and regression.

A. Uncertainty-Aware Sampling for Proposal Network

Prevailing sampling strategies struggle in providing adequate positive samples for size-limited instances, and the core issue lies in that they mandate severe requirement on the priors of being a positive sample [3], [28]. Moreover, the inherent uncertainty associated with small objects injects instability to the learning, as the detectors may allocate futile focus to those highly unrecognizable samples and result in a nontrivial solution [51], thereby failing to generate sufficient high-quality proposals [27], [38], [52]. In parallel, prior endeavors [36], [37] have shown that a tailored process for each instance holds the potential for optimal sampling. This enlightens us to develop a dynamic strategy, which ponders the intrinsic property (namely localization uncertainty) into sampling, to unearth potential candidates for instances with high certainty and avoid worthless focuses on those with excessive uncertainty, thereby alleviating the sample scarcity and guarantee the optimal learning of the overall network.

1) *Instance-level Uncertainty Modeling*: It is an ill-posed issue to evaluate the uncertainty for a dataset directly as the corresponding labels remain uncharted. Therefore, we prefer to formulate the uncertainty of positive priors for each instance. To endow the proposal network with the capability to perceive uncertainty, an additional location deviation prediction branch is incorporated into RPN, as shown in Fig. 5. Furthermore, we follow the seminal work [23] to model the box representation as a Gaussian distribution. For a predicted box x_p , it can be formulated as:

$$P_{\theta}(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-x_p)^2}{2\sigma^2}}, \quad (1)$$

where θ indicates the parameters to be optimized, while $\sigma = \{\sigma_t, \sigma_l, \sigma_b, \sigma_r\}$ represents the predicted standard deviation emanated from uncertainty branch for each box side, namely top, left, bottom, and right, which reflects the localization confidence of the model for current box. Essentially, we can also parameterize the ground-truth box x_g to a Gaussian

distribution with σ being $\mathbf{0} = \{0, 0, 0, 0\}$. Following [23], we introduce a tailored uncertainty loss for this branch:

$$\mathcal{L}_{\text{unc}}^{\text{RPN}} = \frac{(x_g - x_p)^2}{2\sigma^2} - \frac{1}{2} \log(\sigma^2). \quad (2)$$

Note that only positive priors anticipate the optimization of uncertainty branch. And we use conventional overlap-based assignment scheme to capture positive candidates as follows:

$$\text{IoU}(x_g, x_p) \geq T_{\text{pos}}, \quad (3)$$

where IoU denotes the Intersection-over-Union operation while T_{pos} represents the threshold of being a positive sample. Note that the uncertainty branch is deliberately not optimized for instances where no prior meets the T_{pos} threshold, as estimating uncertainty from only a single, low-quality prior would risk significant optimization instability. Next we elucidate how to model the uncertainty for each instance from its positive priors. The novel branch predicts the deviation associated with each edge, and we capture the normalized uncertainty for each prior as follows:

$$u_{\text{norm}} = \frac{d_t}{d} \sigma_t + \frac{d_l}{d} \sigma_l + \frac{d_b}{d} \sigma_b + \frac{d_r}{d} \sigma_r, \quad (4)$$

in which $d = d_t + d_l + d_b + d_r$, and $\{d_t, d_l, d_b, d_r\}$ denotes the corresponding Euclidean distance for each edge between the positive prior and its target. Now, let's consider a general case, for a ground-truth object x_g , given its positive prior set $\mathcal{S}_{\text{pos}} = \{x_p^i\}_{i=1,2,\dots,N}$, then the instance-level uncertainty is formulated as follows:

$$u_g = \frac{\sum_{i=1}^N \Gamma(\delta_i) \cdot u_{\text{norm}}^i}{\sum_{i=1}^N \Gamma(\delta_i)}, \quad (5)$$

where u_{norm}^i denotes normalized uncertainty for each i -th prior namely x_p^i and δ_i indicates the IoU of current prior to its GT target, while i indexes the priors. Moreover, $\Gamma(\cdot)$ is a non-linear transformation:

$$\Gamma(z) = \frac{1}{1 + \exp(-2z)}. \quad (6)$$

From Eq. 5, the uncertainty u_g for the instance g relies on the ambiguity prediction of its priors and the corresponding IoUs. And we perform a transformation to the IoUs instead of directly utilizing them via Eq. 6, which boosts the weights of those priors that are less aligned with the instance during uncertainty measurement.

2) *Uncertainty-Aware Sampling*: We employ the instance-level uncertainty as a surrogate criterion for assignment:

$$\text{IoU}(\mathbf{x}_g, \mathbf{x}_p) \geq u_g. \quad (7)$$

Priors that meet the above criterion will be optimized via the regression loss $\mathcal{L}_{\text{reg}}^{\text{RPN}}$, and note that the positives priors used for uncertainty modeling will not be involved to this process. Then the objective function for proposal networks can be formulated as follows:

$$\mathcal{L}^{\text{RPN}} = \mathcal{L}_{\text{cls}}^{\text{RPN}} + \mathcal{L}_{\text{reg}}^{\text{RPN}} + \alpha \cdot \mathcal{L}_{\text{unc}}^{\text{RPN}}. \quad (8)$$

We take conventional Cross Entropy loss and Smooth L1 loss for $\mathcal{L}_{\text{cls}}^{\text{RPN}}$ and $\mathcal{L}_{\text{reg}}^{\text{RPN}}$, respectively. Moreover, following the setup in MMDetection [53] and previous works [16], [28], [54], the loss weights for $\mathcal{L}_{\text{cls}}^{\text{RPN}}$ and $\mathcal{L}_{\text{reg}}^{\text{RPN}}$ are set to 1 by default, while the term α scales the contribution of uncertainty-modeling branch during the optimization process.

This uncertainty-aware sampling strategy mines potential candidates in a dynamic fashion, as the uncertainty prediction is evolving. To be specific, the novel branch tends to predict large uncertainty for all the priors at early stage. With the training progresses, the model adeptly manages the localization distribution of those instances with proper characteristics, and the above solution genuinely eases the stringent requirements mandated by conventional assignment strategies in a customized manner. Meanwhile, there indeed exists a vast collection of instances whose uncertainty predictions remain in a high numerical range even at the end of training. At this point, the scheme in Eq. 7 will essentially degenerate into the common scenario, as the estimated uncertainty u_g exceeds the predefined positive threshold T_{pos} , and we will provide the uncertainty distribution in the experimental section to intuitively illustrate these patterns. By this setup, we can address the ill-posed issue of assessing uncertainty for instances, which will be leveraged to explore potential samples for proposal network training while refraining from introducing low-quality priors, thereby preventing the model from falling into trivial solutions. Moreover, the whole procedure is hyperparameter-free.

B. Perception-and-Interaction

Seminal works [10], [11], [17] usually employ a size-dependent strategy to extract regional features for proposals. Take MMDetection [53], one of the most prevailing detection toolboxes as an example, it decides the pyramid level to conduct RoI Align as follows:

$$l = \lfloor \sqrt{w \cdot h} / s_{\text{base}} + \epsilon \rfloor, \quad (9)$$

where w and h represent the size of current proposal, while s_{base} denotes the base scale and is default to 56 in [53], and the term ϵ is for numerical stability. In this way, the proposals at various scales could be assigned to appropriate pyramids, guaranteeing the explicit and simultaneous optimization of the

hierarchy. Unfortunately, small objects, within the area range of 32×32 for prevailing SOD datasets [2], [20], [55], are usually allocated to lowest level namely P_2 according to Eq. 9. In other words, the higher pyramids can only be updated implicitly through the top-down modulation in FPN (Feature Pyramid Network) [17], [56]. Worse still, the features used for detection at two stages often exhibit inconsistency as shown in Fig. 3. This essentially exposes a perplexing yet underexplored issue in SOD task: which layer should we register to derive powerful representation for small instances?

To alleviate this hierarchy uncertainty, we exploit off-the-shelf cues to construct the Perception-and-Interaction scheme, thereby procuring the robust and discriminative region features. Our main consideration is straightforward: why discarding the features that are competent to yield high-quality proposals in the first stage, notably in light of the limited availability of inherent information for size-limited instances? In Fig. 6, for each proposal emanated from RPN, the pipeline concurrently extracts region features from two levels: the anchor-based level from which the proposal originates, and the hierarchy assigned by the heuristic size-dependent mapping strategy (e.g., Eq. 9). Subsequently, mutual perception and cross interaction are employed to facilitate the final representation. Next we illuminate the details of this strategy.

1) *Reciprocal Perception*: For any proposal, the regional features pooled from the original pyramid and the assigned one are denoted as $\mathbf{F}_o$ and $\mathbf{F}_a$, respectively, each with dimensions $H \times W \times 256$. Despite originating from the same region, $\mathbf{F}_o$ and $\mathbf{F}_a$ essentially capture object representations from different aspects, thereby establishing a complementary relationship [29], [57]. Specifically, $\mathbf{F}_o$ encompasses highly compact patterns and exhibits a pronounced homogenization since the elements are pooled from a relatively narrow region, while $\mathbf{F}_a$ retains abundant details entailed for perceiving small instances. Therefore, we investigate a mutual perception scheme to leverage the structure information in $\mathbf{F}_o$ to enrich the spatial representation in original pyramid, namely *Analytic Perception*. In parallel, the high compactness inherent in $\mathbf{F}_a$ endows the assigned pyramid with resilience to variations, which is dubbed as *Holistic Perception*.

To furnish the spatial details of original pyramid, a natural idea is incorporating $\mathbf{F}_a$ to $\mathbf{F}_o$. Nevertheless, for small instances, the pooling operation inevitably induces distortion which further impedes the final representation quality [58]. Hence, the input channels are first compressed to 128 to filter the negative impact of cross-level artifacts, and moreover, minimize the computational overhead. And we denote these features as $\tilde{\mathbf{F}}_a$ and $\tilde{\mathbf{F}}_o$ for simplicity. Then, to fully exploit the structural patterns and further empower the spatial sensitivity, enlightened by [59], we leverage $\tilde{\mathbf{F}}_a$ to generate a series of position-dependent kernels, which will then be correlated with $\tilde{\mathbf{F}}_o$ to recover its layouts as follows:

$$\mathbf{F}_{\text{ap}} = \mathbf{W}_{\text{ap}} \otimes \tilde{\mathbf{F}}_o, \quad (10)$$

where $\mathbf{F}_{\text{ap}}$ denotes the features after Analytic Perception and $\otimes$ is the convolution operation. $\mathbf{W}_{\text{ap}} \in \mathbb{R}^{H \times W \times K^2}$ indicates the generated kernels which are computed as follows:

$$\mathbf{W}_{\text{ap}} = \text{Norm}(\varphi_{\text{ap}}(\tilde{\mathbf{F}}_a)), \quad (11)$$

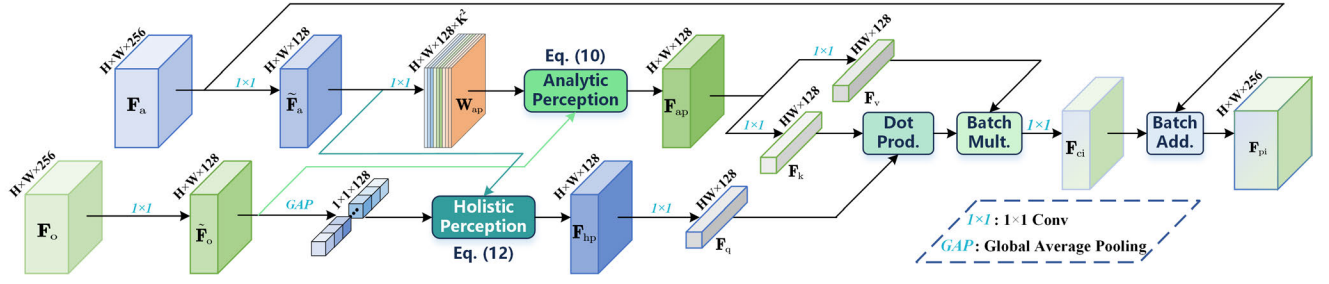


Fig. 6. Illustration of the proposed Perception-and-Interaction strategy. The regional representations pooled from original level ($\mathbf{F}_o$) and assigned level ($\mathbf{F}_a$) are first reduced in dimensionality from 256 to 128 for lower complexity, producing $\tilde{\mathbf{F}}_o$ and $\tilde{\mathbf{F}}_a$, respectively. These two embeddings are utilized to derive kernels for subsequent mutual perceptions—the Analytic Perception (Eq. 10) and Holistic Perception (Eq. 12)—thereby yielding $\mathbf{F}_{ap}$ and $\mathbf{F}_{hp}$. Finally, cross-level interaction is employed to acquire the final region representations, denoted by $\mathbf{F}_{pi}$.

where φ_{ap} represents 1×1 convolution to generate position-dependent kernels. Then, following pioneering work [59], we use Norm to normalize the kernels and maintain the magnitude of input features. For Holistic Perception branch, our main idea lies in that utilizing the highly compact information to impart the assigned hierarchy (namely the lowest-level pyramid) with robustness against deformation and noise. To this end, we simply use Global Average Pooling (GAP) to provide a holistic view of $\tilde{\mathbf{F}}_o$, and this procedure can be formulated as follows:

$$\mathbf{F}_{hp} = \text{GAP}(\tilde{\mathbf{F}}_o) \otimes \tilde{\mathbf{F}}_a, \quad (12)$$

in which $\mathbf{F}_{hp}$ indicates the representations processed by Holistic Perception, while GAP denotes the Global Average Pooling. By the devised reciprocal paradigm, the characteristic inherent in $\tilde{\mathbf{F}}_a$ and $\tilde{\mathbf{F}}_o$ is preserved while deficiencies in detail description and robustness to noise are complemented. More importantly, such bidirectional perception alleviates the misalignment between two distinct pyramids, facilitating subsequent interaction.

2) *Cross Interaction*: Given $\mathbf{F}_{ap}$ and $\mathbf{F}_{hp}$, our aim is to calibrate these region features from two pyramids, and then derive a powerful representations with discriminative patterns. To this end, we further develop a interaction scheme to thoroughly leverage the intrinsic correlations between the regional features at two hierarchies, which draws inspiration from substantial successes of self-attention mechanism [72] and can be described as follows:

$$\mathbf{F}_{ci} = \text{softmax} \left(\frac{\mathbf{F}_q \cdot \mathbf{F}_k^T}{\sqrt{d}} \right) \cdot \mathbf{F}_v, \quad (13)$$

where softmax and $\mathbf{F}_{ci}$ denote the Softmax function and the region representations emanated from Cross Interaction scheme, respectively. $\mathbf{F}_q$, $\mathbf{F}_k$, and $\mathbf{F}_v$ illustrate the embedding vectors which can be captured as follows:

$$\mathbf{F}_q = \varphi_q(\mathbf{F}_{hp}), \mathbf{F}_k = \varphi_k(\mathbf{F}_{ap}), \mathbf{F}_v = \varphi_v(\mathbf{F}_{ap}), \quad (14)$$

where φ_q , φ_k , and φ_v symbolize the embedding convolutions. Moreover, d signifies the intermediate dimension and is set to 128 by default. Note that unlike the conventional approaches, our correlation performs in a cross-level fashion. Henceforth, the input to the embedding convolutions have various choices and we conduct ablation studies to seek the optimal setting. Then the region representations $\mathbf{F}_{pi}$ are as follows:

$$\mathbf{F}_{pi} = \varphi_{re}(\mathbf{F}_{ci}) + \mathbf{F}_a, \quad (15)$$

where φ_{re} restores the output channels to 256, and the incorporation of $\mathbf{F}_a$ is expected to underscore the inherent structural cues in low-level features.

IV. EXPERIMENTS

A. Experimental Setups

1) *Datasets*: We take two large-scale benchmarks tailored for small object detection, namely SODA-D and SODA-A [2], along with three publicly SOD-related datasets, including COCO [22], Tsinghua-Tencent 100K [73], and VisDrone [74], to validate the effectiveness of our uncertainty learning framework for SOD task.

a) *SODA-D*: SODA-D [2] is a large-scale dataset for small object detection in traffic scenario, which contains 24,828 high-resolution images. SODA-D annotates 278,433 instances covering nine categories in driving scene: *people*, *rider*, *bicycle*, *motor*, *vehicle*, *traffic-sign*, *traffic-light*, *traffic-camera*, and *warning-cone*. Moreover, as a challenging dataset for small object detection, SODA-D meticulously incorporates 153,976 *ignore* annotations which aids the model in focusing on the size-limited instances. SODA-D exhibits rich diversity in terms of shooting view, location, weather, to name a few, rendering it more closely aligned with real-world applications.

b) *SODA-A*: SODA-A [2] comprises 2513 high-quality images from Google Earth, on which 872,069 instances are labeled with oriented bounding boxes and span over nine classes: *airplane*, *helicopter*, *small-vehicle*, *large-vehicle*, *ship*, *container*, *storage-tank*, *swimming-pool*, and *windmill*. More than 90% of instances in the SODA-A dataset exhibit an area smaller than 1024 pixels, and the average absolute size of them is merely 14.75 pixels. In parallel, the instances in SODA-A can appear in arbitrary orientations and are with significant density variations, posing substantial challenges to the detectors.

c) *COCO*: Microsoft COCO [22] (COCO) is a large-scale dataset consisting of over 123,000 images, with more than 886,000 labeled instances across 80 object categories. It includes a diverse range of images from everyday scenes, offering detailed annotations for detection, segmentation, and captioning tasks. In the past decades, COCO has been the de facto standard for evaluating the performance of various detectors due to its rich variety of object categories, complex scenes, and significant variations in scale and occlusion. Moreover, more than 30% of the instances in COCO have an area

smaller than 1024 pixels, making it a widely used benchmark for evaluating the effectiveness of SOD-oriented approaches.

d) *Tsinghua-tencent 100K*: Tsinghua-Tencent 100K [73] (short for TT100K) is captured from 100,000 Tencent Street View panoramas, providing 30,000 traffic-sign instances with detailed annotations and covering large variations in illuminance and weather conditions. As one of the most frequently used datasets for validating the performance of SOD approaches, TT100K is featured by not only its small-size, but also the rich distribution of categories.

e) *VisDrone*: VisDrone [74] provides 10,209 drone-viewed images, annotated with more than 542,000 instances spanning 10 traffic-related object categories. It encompasses several highly demanding scenarios for object detection, such as small object sizes, heavy occlusion, and densely packed conditions.

2) *Implementation Details*: The `train-set` of the aforementioned five datasets are utilized for model training, while the `test-set` of SODA-D, SODA-A, and TT100K, as well as the `val-set` of COCO and VisDrone, are employed to evaluate and conduct performance comparisons. During pre-processing, we first crop the original images from SODA-D and SODA-A into patches of size 800×800 pixels, which are subsequently resized to 1200×1200 pixels for training and inference. For the TT100K dataset, images are resized to 1024×1024 pixels, and the categories with fewer than 100 instances are excluded, following the conventional setups [73]. For COCO and VisDrone, we adhere to the standard settings employed in previous works [12], [13], [17], [84], [85], maintaining the shortest side of each image to be at least 480 pixels and at most 800 pixels, while ensuring that the longest side does not exceed 1333 pixels. Except for VisDrone, which employs augmentation techniques such as brightness and contrast adjustments consistent with pioneering work [84], all other datasets include random flipping for data augmentation only. All the models are trained on two RTX 3090 GPUs for 12 epochs with a batch size of 4. Without specific description, we use ResNet-50 [86] with FPN [17] to construct hierarchical representations, and the initial learning rate is set to 0.01 and will be decayed by 0.1 after 8 and 11 epoch. Moreover, the default optimizer is SGD with the momentum of 0.9 and the weight decay of 0.0001.

3) *Evaluation Metrics*: We follow the default metrics in [2] and [22] for performance evaluation. To be specific, the mean Average Precision is computed across all categories over 10 IoU thresholds ranging from 0.5 to 0.95 with an interval at 0.05, along with AP_{50} and AP_{75} . In addition, the results at specific area ranges are also investigated, namely AP_{es} , AP_{rs} , AP_{gs} , and AP_N for SODA-D and SODA-A datasets, which corresponds to *extremely Small*: (0, 144], *relatively Small*: (144, 400], *generally Small*: (400, 1024], and *Normal*: (1024, 2000]. For COCO, TT100K, and VisDrone datasets, we report the performance of AP_S , AP_M , and AP_L , which is consistent with previous work [22], [73], [74].

B. Main Results

We evaluate mainstream detectors and Unc-SOD on the large-scale benchmarks SODA-D and SODA-A. Moreover, we also compare the performance between the baseline model and our method on COCO, TT100K, and VisDrone datasets.

1) *SODA-D*: The results on the SODA-D `test-set` are presented in TABLE I. Our Unc-SOD achieves a superior AP of 31.0% under identical training setups, surpassing the baseline [10] by a significant 2.1% margin. Such a substantial performance gain stems from directly addressing two fundamental challenges in small object detection: sample scarcity and hierarchical inconsistency, where previous methods exhibit clear limitations.

First, existing label assignment schemes, from generic approaches [12], [14], [36] to SOD-specific ones [16], [28], [41], struggle to provide sufficient high-quality samples. For instance, RFLA [28] exhibits performance degradation on extremely small objects (-0.6% points), while the coarse-to-fine pipeline in CFNet [16] incurs high computational costs. In contrast, by leveraging instance-level uncertainty, our strategy dynamically unearths qualified candidates with no extra computational overhead, thereby effectively mitigating the critical issue of sample scarcity. As validated in our ablation study (TABLE VI), this designed strategy alone boosts the baseline AP by 1.0% and, more importantly, raises the challenging AP_{es} from 13.8% to 14.3%, offering a more efficient approach to overcoming the sample paucity bottleneck.

Second, our Perception-and-Interaction scheme effectively alleviates the feature inconsistency between the two stages, a longstanding challenge when applying two-stage paradigms to the SOD task. Early attempts to refine proposals like Cascade RPN [27] show minimal gains. Moreover, other SOD-oriented efforts exploring powerful hierarchical representations [62], [71] or reconstruction-based features [31] have also fallen short of delivering substantial improvements. Similarly, the paradigms that model box uncertainty for detection, such as GFL [25] and KL [23], improve localization, but the primary gains come from larger instances, failing to resolve the intrinsic hierarchical ambiguity in SOD. The proposed pipeline, however, is specifically designed for this purpose; it contributes a 1.5% AP gain on its own and is the primary driver behind the remarkable 3.0% improvement in the strict AP_{75} metric (TABLE VI). This highlights its superior ability to generate accurate bounding boxes by aligning features across the detection pipeline.

In a broader context, our framework significantly outperforms other paradigms. Query-based methods [7], [70] and lightweight models [65], [66], [67], [68], [69], despite their success in generic detection, struggle with the unique challenges posed by objects of extreme size. Moreover, unlike the YOLO family which requires complex data augmentations and extended training epochs, Unc-SOD achieves state-of-the-art results on small-object metrics using only a $2\times$ training schedule without additional augmentations. Collectively, these findings reinforce the potential of uncertainty-modeling as a powerful paradigm for small object detection.

2) *SODA-A*: Table II reports the performance of mainstream detection approaches on the SODA-A `test-set`. On this challenging aerial imagery benchmark, our framework achieves an overall AP of 34.8%, setting a new state-of-the-art. Moreover, Unc-SOD significantly outperforms one-stage detectors with their inherent limitations and establishes a clear advantage over other proposal-based competitors.

Specifically, even top-performing rivals like Oriented RCNN [54] and CFNet [16], which both score a strong 34.4% AP,

TABLE I

COMPARISON WITH STATE-OF-THE-ART DETECTION APPROACHES ON THE SODA-D TEST-SET, WHERE ‘BASELINE’ REFERS TO FASTER RCNN [10], SERVING AS THE BASELINE FOR THE TWO-STAGE METHODS IN THE TABLE. ‘SCHEDULE’ DENOTES THE NUMBER OF EPOCHS FOR TRAINING, IN WHICH ‘1×’ CORRESPONDS TO 12 EPOCHS AND ‘50e’ INDICATES 50 EPOCHS. THE FPS RESULTS ARE EVALUATED ON A SINGLE RTX 3090 GPU UNDER THE SAME COMPUTATIONAL LOAD. † INDICATES THAT AUXILIARY AUGMENTATION STRATEGIES (MIXUP, MOSAIC, etc) ARE INCLUDED. * REPRESENTS THAT THE RESULTS ARE SOURCED FROM THE ORIGINAL PAPER

Method	Publication	Schedule	AP	AP ₅₀	AP ₇₅	AP _{eS}	AP _{rS}	AP _{gS}	AP _N	Param(M)	FLOPs(G)	FPS
One-stage												
RetianNet [12]	ICCV’17	1×	28.2	57.6	23.7	11.9	25.2	34.1	44.2	35.7	299.5	21.1
FCOS [13]	ICCV’19	1×	23.9	49.5	19.9	6.9	19.4	30.9	40.9	31.7	284.5	20.8
ATSS [36]	CVPR’20	1×	26.8	55.6	22.1	11.7	23.9	32.2	41.3	31.3	290.8	19.5
YOLOX [60]	ArXiv’21	70e	26.7	53.4	23.0	13.6	25.1	30.9	30.4	8.9	48.1	43.6
DyHead [61]	CVPR’21	1×	27.5	56.1	23.2	12.4	24.4	33.0	41.9	38.2	159.6	12.8
GFL [25]	NeurIPS’20	1×	29.0	57.3	25.2	12.8	25.4	35.1	44.2	31.5	295.3	19.3
CFPT [62]	TGRS’25	1×	29.2	58.5	25.0	13.3	26.2	35.0	43.7	37.3	302.2	17.5
KLDet* [63]	TGRS’25	1×	25.9	53.8	21.4	10.7	22.2	31.9	41.6	31.9	284.5	20.0
Keypoint-based												
CornerNet [15]	ECCV’18	2×	24.6	49.5	21.7	6.5	20.5	32.2	43.8	201.0	1104.1	6.7
CenterNet [14]	ArXiv’19	70e	21.5	48.8	15.6	5.1	16.2	29.6	42.4	70.8	137.2	34.3
RepPoints [64]	ICCV’19	1×	28.0	55.6	24.7	10.1	23.8	35.1	45.3	36.6	274.0	18.2
Lightweight												
YOLOv5-M† [65]	GitHub’20	3×	30.6	58.4	27.6	10.3	25.3	38.7	49.3	25.1	64.4	185.2
YOLOv8-M† [66]	GitHub’23	3×	31.3	59.2	28.3	10.8	26.0	39.5	50.3	25.9	79.1	161.3
YOLOv10-M† [67]	NeurIPS’24	3×	31.0	58.9	28.1	11.3	25.7	39.1	49.4	16.5	64.0	163.9
YOLO11-M† [68]	GitHub’24	3×	31.5	59.5	28.5	11.4	26.6	39.6	49.3	20.3	68.2	158.7
RT-DETR-L† [69]	CVPR’24	3×	22.1	52.6	14.4	5.4	16.5	29.4	41.3	32.0	103.5	125.0
Query-based												
Deformable-DETR [7]	ICLR’20	50e	19.2	44.8	13.7	6.3	15.4	24.9	34.2	35.2	739.1	18.5
Sparse RCNN [70]	CVPR’21	1×	24.2	50.3	20.3	8.8	20.4	30.2	39.4	106.0	213.0	17.6
Two-stage												
Baseline [10]	NeurIPS’15	1×	28.9	59.4	24.1	13.8	25.7	34.5	43.0	41.2	292.3	20.0
Cascade RPN [27]	NeurIPS’19	1×	29.1	56.5	25.9	12.5	25.5	35.4	44.7	41.8	291.9	16.6
KL [23]	CVPR’19	1×	29.4	59.2	24.7	14.0	26.3	36.0	44.2	41.2	292.1	17.6
RFLA [28]	ECCV’22	1×	29.7	60.2	25.2	13.2	26.9	35.4	44.6	41.2	292.1	19.2
CFINet [16]	ICCV’23	1×	30.7	60.8	26.7	14.7	27.8	36.4	44.6	44.0	312.6	16.8
SR-TOD [31]	ECCV’24	1×	29.3	60.0	24.5	13.8	26.0	35.6	43.4	42.2	292.1	18.9
SimD [41]	IROS’24	1×	27.8	58.0	22.8	12.2	24.6	33.6	41.9	41.2	292.1	19.4
BAFNet [71]	TGRS’25	1×	30.1	61.0	25.5	14.5	26.6	36.2	44.7	51.0	1068.6	8.6
Unc-SOD (ours)	-	1×	31.0	60.8	27.1	14.9	27.6	36.9	45.8	41.5	298.5	19.1
Unc-SOD (ours)	-	2×	32.5	62.5	28.7	16.4	29.1	38.6	46.9	41.5	298.5	19.1

TABLE II

COMPARISON WITH STATE-OF-THE-ART DETECTION APPROACHES ON THE SODA-A TEST-SET, WHERE ‘BASELINE’ REFERS TO ROTATED FASTER RCNN [10], SERVING AS THE BASELINE FOR THE TWO-STAGE METHODS IN THE TABLE. OTHER SETTINGS ARE CONSISTENT WITH TABLE I. THE FPS RESULTS ARE EVALUATED ON A SINGLE RTX 3090 GPU UNDER THE SAME COMPUTATIONAL LOAD.

* REPRESENTS THAT THE RESULTS ARE SOURCED FROM THE ORIGINAL PAPER, WHERE THE INPUT RESOLUTION FOR TRAINING AND INFERENCE IS 800×800 PIXELS

Method	Publication	Schedule	AP	AP ₅₀	AP ₇₅	AP _{eS}	AP _{rS}	AP _{gS}	AP _N	Param(M)	FLOPs(G)	FPS
One-stage												
Rotated RetinaNet [12]	ICCV’17	1×	26.8	63.4	16.2	9.1	22.0	35.4	28.2	26.2	800.2	18.3
S ² A-Net [75]	TGRS’22	1×	28.3	69.6	13.1	10.2	22.8	35.8	29.5	38.6	732.7	16.7
Oriented RepPoints [76]	CVPR’22	1×	26.3	58.8	19.0	9.4	22.6	32.4	28.5	55.7	827.2	16.5
DHRec [77]	TPAMI’22	1×	30.1	68.8	19.8	10.6	24.6	40.3	34.6	32.0	792.8	17.2
CFPT [62]	TGRS’25	1×	25.9	63.3	14.3	9.0	21.2	34.6	27.9	37.1	296.9	16.5
Two-stage												
Baseline [10]	NeurIPS’15	1×	32.5	70.1	24.3	11.9	27.3	42.2	34.4	41.1	292.2	18.6
Gliding Vertex [78]	TPAMI’21	1×	31.7	70.8	22.6	11.7	27.0	41.1	33.8	41.1	292.3	17.8
Oriented RCNN [54]	ICCV’21	1×	34.4	70.7	28.6	12.5	28.6	44.5	36.7	41.1	292.4	17.9
DODet [79]	TGRS’22	1×	31.6	68.1	23.4	11.3	26.3	41.0	33.5	69.3	555.5	15.5
CFINet [16]	ICCV’23	1×	34.4	73.1	26.1	13.5	29.3	44.0	35.9	44.0	312.6	17.4
MENet* [80]	TGRS’24	1×	32.1	-	-	10.8	26.3	41.4	-	-	-	-
SR-TOD [31]	ECCV’24	1×	32.7	70.5	24.1	12.2	27.5	42.6	34.5	42.2	292.1	17.3
STD [81]	AAAI’24	1×	30.3	67.4	21.2	12.4	26.6	36.7	30.7	30.2	473.8	6.0
Unc-SOD (ours)	-	1×	34.8	73.6	26.4	13.8	29.7	44.7	36.5	41.5	298.5	17.6

show different specializations. While Oriented RCNN excels at high-quality localization, achieving a leading AP₇₅ of 28.6% at the cost of subpar performance on extremely small objects (AP_{eS} of 12.5%), CFINet is more adept at handling such

size-limited objects (AP_{eS} of 13.5%). Furthermore, other recent efforts, whether exploring SOD-specific modifications such as MENet [80] and SR-TOD [31], novel box representations like DODet [79] and Gliding Vertex [78], or leveraging

TABLE III

PERFORMANCE COMPARISON ON THE COCO *val*-SET AMONG GENERIC DETECTORS, SOD-ORIENTED APPROACHES AND OUR SOLUTION, WITH ALL MODELS TRAINED FOR 12 EPOCHS. NOTE † INDICATES THE TRICKS IN [53] ARE USED AND * REPRESENTS THAT THE RESULTS ARE SOURCED FROM THE ORIGINAL PAPER

Method	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Generic Object Detection						
RetinaNet [12]	36.6	55.7	38.8	21.6	40.3	47.9
FCOS† [13]	38.5	57.8	41.5	22.8	42.6	49.5
Faster RCNN [10]	37.2	58.1	40.3	21.0	40.8	48.4
Small Object Detection						
CFPT [62]	37.2	58.0	39.6	21.7	41.0	48.5
KLDet [63]	35.7	56.3	38.1	21.5	39.0	44.4
EFPN* [18]	38.2	-	-	22.7	41.0	49.4
RFLA [28]	37.7	58.7	40.7	20.7	40.8	50.0
BAFNet [71]	37.7	58.6	40.8	22.7	41.2	48.2
CFINet [16]	38.1	58.8	41.3	22.6	42.3	47.9
SimD [41]	35.7	56.3	38.7	20.8	39.8	45.7
Unc-SOD (ours)	38.6	59.3	42.1	23.3	42.5	49.0

TABLE IV

PERFORMANCE COMPARISON ON THE TT100K *test*-SET AMONG GENERIC DETECTORS, SOD-ORIENTED APPROACHES AND OUR SOLUTION. OTHER SETTINGS ARE CONSISTENT WITH TABLE III

Method	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Generic Object Detection						
Faster RCNN [10]	62.5	82.2	74.7	39.7	72.0	81.2
FCOS [13]	55.0	72.4	64.1	29.2	64.7	78.7
Small Object Detection						
CFPT [62]	56.9	74.4	65.8	29.3	67.8	80.4
KLDet [63]	56.6	75.2	66.3	33.9	66.6	73.2
RFLA [28]	62.3	82.0	73.7	40.7	71.3	80.9
CFINet [16]	62.9	82.4	75.1	39.5	72.5	81.2
SimD [41]	61.9	81.1	73.4	38.0	71.8	80.5
BAFNet [71]	62.9	82.7	74.7	40.9	72.1	80.8
Unc-SOD (ours)	64.2	84.1	76.3	41.9	73.4	81.4

powerful ViTs [87] for oriented object detection [81], fail to deliver significant breakthroughs on size-limited instances.

In contrast, benefiting from the Uncertainty-aware Sampling strategy that mines sufficient qualified samples and the Perception-and-Interaction pipeline that yields more discriminative representations, we establish a more holistically powerful solution, thereby achieving the best AP_S of 13.8% at a competitive 17.6 FPS. This further demonstrates the flexibility and effectiveness of our uncertainty-modeling approach in handling the complex challenges of detecting small, oriented objects in aerial scenes.

3) *COCO*: On the COCO *val*-set, our Unc-SOD demonstrates its targeted effectiveness and robust performance, as shown in TABLE III. The core strength of our framework is most evident in detecting small objects, where it achieves a top-performing AP_S of 23.3%. This result represents a substantial 2.3% gain over the baseline (21.0%) and decisively surpasses a spectrum of SOD-oriented competitors. Notably, while top performers like EFPN [18] (22.7%) and CFINet [16] (22.6%) show modest gains, other specialized methods such as RFLA [28] (20.7%) and SimD [41] (20.8%) even fail to match the generic baseline's performance, underscoring the challenge of generalizing SOD designs.

This decisive advantage on size-limited instances translates into the best performance among all the methods. Our

TABLE V

PERFORMANCE COMPARISON ON THE VisDrone *val*-SET AMONG GENERIC DETECTORS, SOD-ORIENTED APPROACHES AND OUR SOLUTION. EACH MODEL IS TRAINED FOR 15 EPOCHS, DURING WHICH THE INITIAL LEARNING RATE IS DECREASED BY A FACTOR OF 10 FOLLOWING THE 11TH AND 14TH EPOCHS.

* DENOTES THAT THE RESULTS ARE SOURCED FROM THE ORIGINAL PAPER, WHILE CLUSDET [82] AND HAWKNET [83] ARE TRAINED FOR 42 EPOCHS AND 36 EPOCHS, RESPECTIVELY

Method	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Generic Object Detection						
RetinaNet [12]	20.2	36.9	19.5	-	-	-
FCOS [13]	21.5	36.6	21.8	12.9	32.0	35.4
ATSS [36]	22.7	40.0	22.5	15.8	31.2	32.5
Faster RCNN [10]	26.3	43.8	27.1	17.4	37.8	38.8
ClusDet* [82]	26.7	50.6	24.7	17.6	38.9	51.4
HawkNet* [83]	25.6	44.3	25.8	19.9	36.0	39.1
Small Object Detection						
CFPT [62]	21.1	37.6	20.6	10.7	35.2	40.3
KLDet [63]	20.5	35.4	20.8	12.8	30.9	32.7
RFLA [28]	26.7	44.7	27.8	18.2	38.1	38.7
CFINet [84]	27.3	46.4	28.0	19.4	37.5	38.8
SimD* [41]	26.5	47.9	-	-	-	-
BAFNet [71]	27.0	44.9	27.9	17.5	38.2	39.4
SR-TOD [31]	26.5	44.2	27.6	17.6	38.3	39.1
Unc-SOD (ours)	27.8	47.3	28.1	20.0	38.2	39.0

Unc-SOD obtains an overall AP of 38.6%, outperforming both the baseline by 1.4% points and strong SOD-specific competitors like EFPN [18] (38.2%) and CFINet [16] (38.1%). Ultimately, the strong performance on the general-purpose benchmark COCO validates that our uncertainty-driven designs provide a robust advantage for small object detection, rather than being a niche optimization for specific datasets.

4) *TT100K*: We also evaluate our method on the commonly used SOD dataset TT100K *test*-set, and the results are presented in TABLE IV. Unc-SOD achieves consistent improvements, with AP_S increasing from 39.7% to 41.9%, outperforming the second-best approach BAFNet [71] by 1.0% points. In parallel, the AP_L remains virtually unchanged, and we conjecture that the uncertainty we discussed poses a negligible influence on large instances. Generally, our framework holds a clear advantage over other SOD-specific competitors, showcasing its generality in diverse, real-world scenes.

5) *VisDrone*: The experiments on the VisDrone *val*-set demonstrate consistent improvements of our approach over the baseline, as illustrated in TABLE V. The key to this success lies in its superior ability to detect small instances in complex aerial scenes. Unc-SOD achieves a top-performing AP_S of 20.0%, which not only outperforms the drone-centric approaches like HawkNet [83] (19.9%) and ClusDet [82] (17.6%) that are trained on longer schedules, but also leads other SOD-tailored approaches such as CFINet [16] (19.4%) and RFLA [28] (18.2%). This robust performance on the VisDrone benchmark unveils the potential and generality of our uncertainty learning framework in drone-view scenarios.

C. Ablation Studies

This section we perform a series of ablation experiments to attest the efficacy of designed components and the optimal choices for several hyperparameters and structures. In parallel, the constructive discussions about these results are

TABLE VI

ABLATIONS OF OUR APPROACH, WHERE BASELINE INDICATES FASTER RCNN WHILE US. AND PI. REPRESENT THE DESIGNED UNCERTAINTY-AWARE SAMPLING STRATEGY AND PERCEPTION-AND-INTERACTION SCHEME, RESPECTIVELY

Dataset			SODA-D				COCO	
Baseline	US.	PI.	AP	AP _{eS}	AP _{rS}	AP _{gS}	AP	AP _S
✓	✗	✗	28.9	13.8	25.7	34.5	37.2	21.0
✓	✓	✗	29.9	14.3	27.0	35.5	38.0	22.4
✓	✗	✓	30.4	14.6	27.0	36.1	38.2	22.7
✓	✓	✓	31.0	14.9	27.6	36.9	38.6	23.3

TABLE VII

ABLATION ANALYSIS OF THE UNCERTAINTY-AWARE SAMPLING STRATEGY, IN WHICH UNC. DENOTES THE AUXILIARY UNCERTAINTY BRANCH ADDED TO VANILLA RPN, WHILE SAM. REFERS TO THE NOVEL LABEL ASSIGNMENT SCHEME

Baseline	Unc.	Sam.	AP	AP _{eS}	AP _{rS}	AP _{gS}
✓	✗	✗	28.9	13.8	25.7	34.5
✓	✓	✗	29.2	13.9	26.2	34.8
✓	✓	✓	29.9	14.3	27.0	35.5

synchronously provided. Note that the experiments in this part are based on the SODA-D test-set.

1) *Investigation of Designed Components:* in TABLE VI, we first ablate each component involved in Unc-SOD to explore their effects to the final performance. It can be seen that introducing the Uncertainty-aware Sampling (US.) strategy and the Perception-and-Interaction (PI.) structure both facilitate the performance, a trend that holds true across both SODA-D and COCO. On SODA-D, US. and PI. individually improve the AP from 28.9% to 29.9% and 30.4%, respectively. This strong consistency is mirrored on the COCO val-set, where they deliver significant improvements on the AP_S metric (22.4% and 22.7% respectively, vs. 21.0% baseline). This demonstrates that sample scarcity and hierarchical feature inconsistency are fundamental challenges for small object detection, not problems unique to a single dataset. Not surprisingly, the incorporation of these two parts contributes to the best results, since Uncertainty-aware Sampling unearths potential candidates for RPN, thereby promoting the quality and quantity of generated proposals. Henceforth, these adequate high-quality proposals further enable the explicit optimization of pyramidal hierarchies, and benefit the Perception-and-Interaction scheme in mitigating the feature inconsistency of the detection between two phases.

2) *Investigation of Uncertainty-Aware Sampling:* The proposed assignment scheme in Sec. III-A involves two major components: the uncertainty-modeling branch and the novel sampling strategy, and we conduct comprehensive ablations to better evaluate the impact of designed parts.

a) *Ablation analysis:* As shown in TABLE VII, extending the conventional RPN with auxiliary uncertainty prediction branch exhibits minimal improvement. This may originate from that modeling the standard deviation of priors fail to handle the thorny issue: the sample paucity. Moreover, it is intractable to precisely capture the location distribution of objects in some challenging scenarios, as discussed in [25] and [23]. However, when integrating the predicted ambiguity into

TABLE VIII

THE EFFECT OF T_{pos} TO THE FINAL PERFORMANCE

T_{pos}	AP	AP _{eS}	AP _{rS}	AP _{gS}
0.50	29.4	14.0	26.3	35.0
0.60	29.9	14.3	27.0	35.5
0.70	29.7	14.3	26.5	35.4
0.80	29.5	13.5	25.9	35.6

TABLE IX

THE CHOICE OF α TO THE FINAL PERFORMANCE

α	0.1	0.5	1.0	2.0
AP	30.6	30.8	31.0	30.5

TABLE X

ABLATION OF DIFFERENT STRATEGIES FOR THE TRANSFORMATION FUNCTION $\Gamma(\cdot)$

$\Gamma(\cdot)$	AP	AP _{eS}	AP _{rS}	AP _{gS}
Uniform Weighting	29.6	14.1	26.5	35.5
IoU Weighting	29.4	14.0	26.3	35.0
Sigmoid-like Mapping (Eq. 6)	29.9	14.3	27.0	35.5

sampling procedure, we observe consistent performance gain as the instance-level uncertainty endows the sampling process with dynamic, thereby increasing the number of samples and preventing the optimization from being overwhelmed by the potential low-quality priors of those highly uncertain instances.

b) *The effect of $\Gamma(\cdot)$:* To capture the instance-aware uncertainty, we devise a non-linear transformation (Eq. 6) to scale the contribution of each prior based on its IoU. A straightforward approach for $\Gamma(\cdot)$ is to handle all priors uniformly. As shown in TABLE X, such a strategy is verified to be simple-yet-effective. The more intuitive strategy of linear IoU-weighting proves surprisingly ineffective, yielding only marginal gains compared to the vanilla uncertainty branch (29.4% vs. 29.2%). This failure stems from the fact that the substantial weights of high-IoU priors lead to an inflated instance-level uncertainty, despite their lower intrinsic ambiguity, as discussed and visualized later. Furthermore, such paradox renders the dynamic sampling (Eq. 7) ineffective, causing it to collapse into a conventional scenario and fail to unearth potential high-quality candidates.

The proposed non-linear transformation in Eq. 6 adheres to this monotonic scheme while judiciously increasing the contribution of lower-IoU priors. This design is predicated on the insight that if a model produces confident responses (low uncertainty) even for less-aligned priors, the underlying object is a high-potential instance capable of furnishing more qualified samples for optimization. Moreover, it creates a more accurate uncertainty landscape that remains anchored by high-quality priors. This structure not only ensures a robust and stable estimation but also safeguards the training process from contamination by potential low-quality samples.

c) *Setting of T_{pos} :* As we alluded to before, the Uncertainty-aware Sampling strategy developed in our framework brings no additional hyperparameters. However, we still need to deliberately determine the criterion of being a positive sample, namely T_{pos} in Eq. 3 for optimizing uncertainty prediction branch. The performance of different choices for

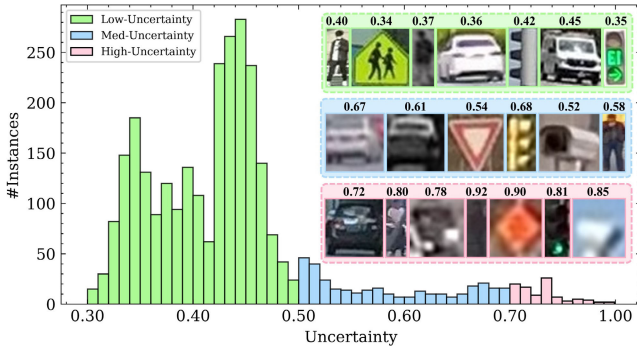


Fig. 7. The uncertainty histogram of 3000 instances randomly captured from SODA-D test-set. We empirically categorize the distribution into three levels—high-, medium-, and low-uncertainty—and representative samples from each subset are exhibited with their corresponding uncertainty annotated above. Note that the instances with uncertainty above than 0.7 are grouped together for clear demonstration.

T_{pos} is illustrated in TABLE VIII, and the top performance is obtained when T_{pos} comes to 0.60. Meanwhile, too rigorous T_{pos} actually impedes the learning of uncertainty branch as there is no adequate samples available, while a lenient setup also fails to model instance-level ambiguity, since the uncertainty of priors positioned distantly from the object region lacks reliability in this context.

d) *Choice of α* : In Eq. 8, the term α controls the loss weight of uncertainty-aware branch, and here we conduct a series of experiments to seek the optimal setup. See TABLE IX, maintaining the loss weight of the uncertainty branch to be consistent with or slightly lower than those of the classification and regression branches in the RPN achieves best performance. Conversely, setting α too low or too high can compromise the final detection accuracy. A low α largely reduces the model to a standard RPN, while a high α places undue emphasis on the uncertainty branch—originally intended as an auxiliary mechanism for procuring potential candidates—thereby overshadowing the primary classification and regression in RPN.

e) *A deeper look into uncertainty-aware sampling*:

In this section, we analyze the uncertainty distribution, and further, investigate the relationship between our proposed dynamic sampling strategy and the conventional IoU-based approach. This allows for a deeper and more intuitive understanding of how our uncertainty-aware framework mitigates the scarcity issue in small object detection. First, we capture the uncertainty distribution of various instances while exhibiting the uncertainty of objects under extreme conditions in Fig. 7. It can be observed that most instances exhibit low uncertainty; however, the objects in severely challenging scenarios (*e.g.*, distortion, extreme illumination, or heavy occlusion) typically show higher uncertainty (*i.e.*, lower localization confidence) due to blurred boundaries and deformed structures, as illustrated by the pink-shaded examples in Fig. 7.

To go a step further, we also investigate the uncertainty distribution across different scale ranges. As demonstrated in Fig. 8, the overall uncertainty pattern for three distinct area subsets—(0, 400], (400, 1024], and (1024, 2000]—is generally consistent with the overall distribution shown in Fig. 7. Next, we delve into each subset to reveal its unique characteristics. For the smallest instances, a considerable number of samples exhibit high uncertainty, corroborating that precisely

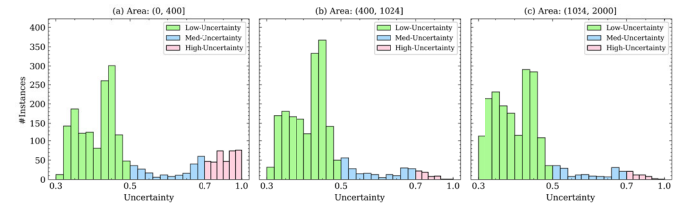


Fig. 8. Uncertainty distributions for different scale ranges, with each distribution derived from 2000 randomly sampled instances from the SODA-D test-set. Similar to Fig. 7, those instances with uncertainty above than 0.7 are grouped together for clear demonstration.

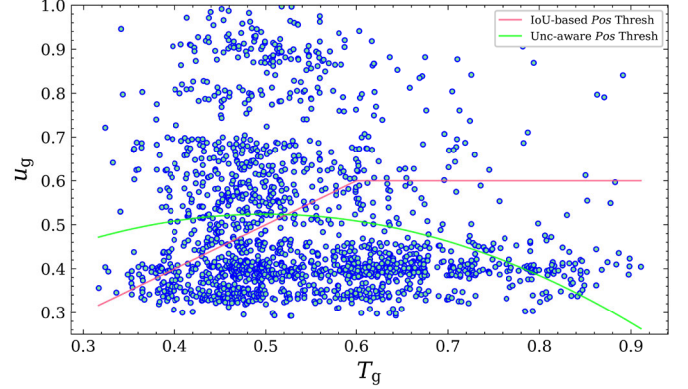


Fig. 9. The distribution of positive threshold between generic IoU-based sampling strategy and our dynamic policy, in which T_g stands for the threshold of being a positive prior for sampled objects in IoU-based assignment scenario and u_g indicates that in our uncertainty-based strategy. The statistical data comprise 2000 instances randomly selected from the SODA-D test-set, and the fitted uncertainty-aware positive threshold function is solely for comparison with the generic IoU-based scheme, not for actual sampling.

determining the boundary of extremely small instances can be inherently intractable for the model. In contrast, the other two plots show a similar trend where the majority of instances are located in the low-uncertainty region. The primary distinction is that larger instances generally possess lower uncertainty and this is especially clear in the (0.3, 0.4) range.

Moreover, to thoroughly assess the effect of uncertainty-aware sampling, we procure the instance-level distribution of positive threshold between generic IoU-based sampling strategy and our dynamic policy. From Fig. 9, those samples with low positive thresholds in IoU-based sampling scheme (say $T_g \leq T_{\text{pos}}$) hold high degree of uncertainty. However, it is also worth noting that a substantial subset of samples within this range exhibit comparatively low uncertainty. Interestingly, the uncertainty-aware positive threshold function—fitted over the sampled points—essentially lies above the IoU-based curve, which suggests that our model effectively addresses the potential underestimation of uncertainty. To go a step further, our uncertainty-aware sampling strategy could dynamically determine the positive criterion on top of predicted localization confidence, thereby unearthing potentially qualified candidates while averting the contamination of low-quality priors.

3) *Investigation of Perception-and-Interaction*: By exploiting the intrinsic correlations inherent in the regional representations from FPN, the Perception-and-Interaction scheme is developed to mitigate the feature inconsistency between two stages induced by the hierarchical uncertainty. Here we

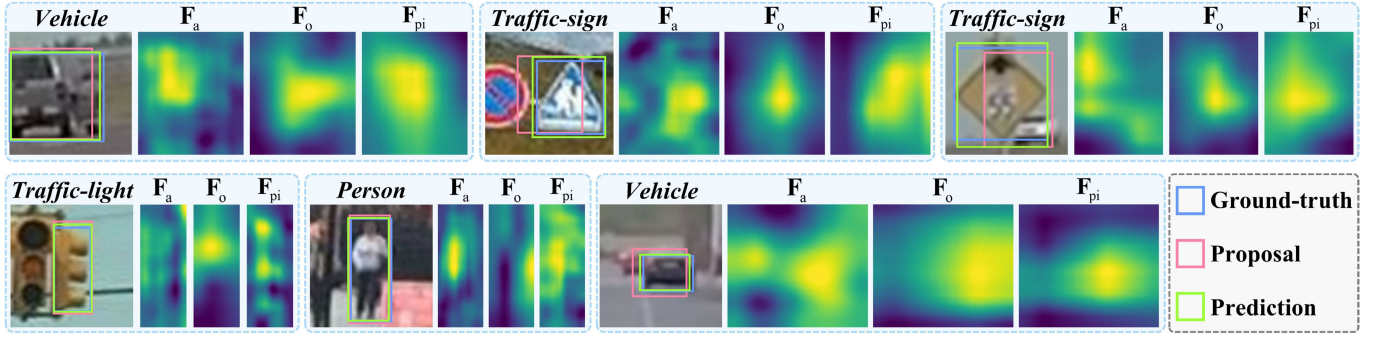


Fig. 10. Visualization of region features from the original pyramid (F_o) and the assigned one (F_a), as well as the final representations emanated from Perception-and-Interaction part namely F_{pi} which embrace more discriminative structures. Each proposal and its corresponding prediction as well as the ground-truth box are highlighted in pink, green and blue, respectively, with the object category indicated above each example.

TABLE XI

THE ABLATIONS ABOUT PERCEPTION-AND-INTERACTION SCHEME. ANA. AND HOL. INDICATE ANALYTIC PERCEPTION AND HOLISTIC PERCEPTION, WHILE INTER. AND ADD. STAND FOR THE INTERACTION AND ADDITION, RESPECTIVELY. HERE THE BASELINE IS FASTER RCNN NAVIGATED BY OUR UNCERTAINTY-AWARE SAMPLING STRATEGY

Perception		Inter.	Add.	AP	AP _{es}	AP _{rs}	AP _{gs}
Ana.	Hol.						
×	×	×	×	29.9	14.3	27.0	35.5
×	×	✓	✓	30.3	14.1	26.7	36.3
✓	×	✓	✓	30.8	14.8	27.6	36.7
×	✓	✓	✓	30.6	14.4	27.3	36.4
✓	✓	✓	×	30.8	14.7	27.4	36.9
✓	✓	✓	✓	31.0	14.9	27.6	36.9

conduct ablations to find the optimal setting and discussions about the structure designs are also provided.

a) *Ablation analysis*: The effects of components in Perception-and-Interaction are shown in TABLE XI. The AP is slightly improved from 29.9% to 30.3% only with the Interaction setup, indicating that bridging the representation gap between the region features emanated from two distinct pyramids is favorable to the subsequent detection. When the region features facilitated by the Analytic Perception or Holistic Perception, further performance gain can be observed. To go a step further, thanks to the structural information and spatial sensitivity conferred by content-aware correlation between the region features from different hierarchies, the integration of Analytic Perception and Interaction achieves impressive results, especially for the challenging AP_{es} and AP_{rs}. And mutual perception, not surprisingly, contributes to the best result, which substantiates that the exploitation of intrinsic relations in hierarchical pyramids is conducive to the discrimination of regional features, as well as compensates the inherently limited information of small objects. Moreover, removing the addition term slightly lowers the overall AP by 0.2% points. This result confirms that while the addition operation conveys the semantic cues inherent in F_a and helps stabilize the training process, the primary contribution comes from our core Perception-and-Interaction pipeline.

To clearly highlight the capability of our Perception-and-Interaction scheme in effectively capturing small instances, we also visualize the region features through the devised

TABLE XII

DIFFERENT INPUT CHOICES FOR INTERACTION MODULE, WHERE Q DENOTES THE SOURCE FEATURE FOR φ_q , AND K&V REPRESENTS THAT FOR φ_k AND φ_v . THE TERM ADD. REFERS TO THE FEATURE FED INTO BATCH ADD. OPERATOR IN EQ. 15. NOTE THAT THE BASELINE IS FASTER RCNN NAVIGATED BY OUR UNCERTAINTY-AWARE SAMPLING STRATEGY

Q	K&V	Add.	AP	AP _{es}	AP _{rs}	AP _{gs}
F_{ap}	F_{hp}	F_o	29.8	14.0	26.9	35.4
F_{hp}	F_{ap}	F_a	31.0	14.9	27.6	36.9
F_{ap}	F_{hp}	F_a	30.5	14.5	27.1	36.5
F_{hp}	F_{ap}	F_o	30.2	14.2	26.7	36.1

pipeline in Fig. 10. Intuitively, the visualization of the fused feature, F_{pi} , demonstrates a significant improvement over the input features, F_a and F_o . It can precisely highlight the object region within the proposal—even when the object occupies only a fraction of the proposal area—while also exhibiting a much clearer structure. This visually confirms that our scheme delivers a highly discriminative region feature. Moreover, benefiting from this enriched representation, the final predicted bounding box could accurately align with the GT instance. This holds true even when the initial proposal is less aligned with the object, as demonstrated by the predictions in Fig. 10.

b) *The input choices for interaction*: In parallel, we investigate the performance of interacting the features from different sources. As exhibited in TABLE XII, the best results can be observed when the query embedding in Eq. 14 is derived from F_{hp} and F_a is involved in the final representation. We conjecture that structural information inherent in low-level pyramids screens the crucial components while empowering the relation modeling between current location and others, thus benefiting the detailed representations entailed for accurate recognition of small instances.

c) *Source level for F_o* : We also explore how the selection of different hierarchies for extracting F_o influences the overall performance, which is demonstrated in TABLE XIV. When delivering region features from P_2 , the mutual interaction essentially degrades into a self-attention mechanism; nevertheless, the resulting accuracy is only 0.2 % points below the optimal configuration. This slight gap can be attributed to the fact that P_2 intrinsically encompasses abundant information pertinent to small objects. In contrast, extracting features



Fig. 11. Qualitative results between Baseline [10] and Unc-SOD on the SODA-D test-set, where our approach is capable of detecting extremely-sized instances (left two columns) while simultaneously suppressing false positives (right two columns), with significant differences highlighted by dashed ellipses.

TABLE XIII

THE INVESTIGATION ABOUT THE DESIGN OF THE PROPOSED PERCEPTION-AND-INTERACTION

Strategy	AP	AP _{eS}	AP _{rS}	AP _{gS}
Baseline	29.9	14.3	27.0	35.5
Addition with Convs	30.3	13.8	27.2	36.2
Concatenation with Convs	30.1	14.3	26.3	36.4
Cross-attention	30.4	14.5	27.3	36.1
Perception-and-Interaction	31.0	14.9	27.6	36.9

TABLE XIV

THE INFLUENCE OF THE ORIGIN OF $\mathbf{F}_0$ ON FINAL PERFORMANCE, IN WHICH ‘ORIGINAL’ REFERS TO THAT POOLING REGIONAL FEATURES FROM THE CORRESPONDING PYRAMIDAL LEVEL ASSOCIATED WITH THE ORIGINAL ANCHOR OF CURRENT PROPOSAL

Source Level	P_2	P_3	P_4	P_5	Original
AP	30.8	30.7	30.7	30.5	31.0

TABLE XV

THE EFFECT OF K TO THE FINAL PERFORMANCE

K	1	3	5
AP	30.6	31.0	30.9

from higher pyramid levels leads to a gradual performance decline, mainly owing to the pronounced homogeneity of these features. For instance, at P_5 with a down-sampling rate of 32, a proposal region of 32×32 actually originates from an extremely narrow spatial area. Furthermore, by leveraging off-the-shelf cues to reconcile the representation discrepancies between the two-stage classification and regression, our framework secures the top performance overall.

d) The exploration of K : In Analytic Perception part, we leverage the low-level feature $\mathbf{F}_a$ to procure content-aware kernels, thereby reconstructing the structure patterns of high-level representation $\mathbf{F}_o$. This involves a hyperparameter namely the kernel size K and we explore the performance of different setups. From TABLE XV, the AP shows discernible disparities when K comes to 3 or 5. However, it drops for $K = 1$ as smaller kernels struggling in capturing the structure patterns in a local region and are inclined to be more sensitive to variations. Therefore, K is set to be 3 in our experiments to strike a balance between computation and accuracy.

TABLE XVI

GENERALIZATION PERFORMANCE OF UNC-SOD ON TWO-STAGE DETECTORS ON THE SODA-D DATASET

Method	AP	AP ₅₀	AP ₇₅	AP _{eS}	AP _{rS}	AP _{gS}	AP _N
Cascade RCNN [11]	31.2	59.9	27.8	14.1	27.5	37.1	46.9
+ Unc-SOD	32.6	61.8	29.3	15.6	29.3	38.9	48.1
DetectoRS [88]	31.3	62.3	26.7	14.3	27.3	37.7	46.7
+ Unc-SOD	32.8	64.0	28.1	15.9	28.9	39.4	48.2

e) The investigation about the structural design: Considering the semantic gap between the region representations from distinct levels, we devise the Perception-and-Interaction pipeline. To thoroughly evaluate our design, we explore several other choices in TABLE XIII, where *Baseline* means Faster RCNN with the proposed uncertainty-aware sampling, while *Addition with Convs* and *Concatenation with Convs* indicate that adding or concatenating $\mathbf{F}_o$ and $\mathbf{F}_a$ followed by a few learnable convolutional layers. Regarding *Cross-attention*, we follow the optimal setups in TABLE XII, namely, using $\mathbf{F}_o$ as the query and $\mathbf{F}_a$ for both the key and value.

It can be seen that naive fusion mechanisms, such as direct addition or concatenation followed by convolution layers, yield only marginal gains. This can be attributed to the substantial semantic gap inherent between representations from two distinct pyramidal levels. In parallel, performing cross-attention between the two-sourced inputs $\mathbf{F}_o$ and $\mathbf{F}_a$ still fails to produce a satisfactory outcome. As discussed in Sec. III-B, the region feature from high-level pyramids may exhibit a pronounced homogenization since the elements are pooled from a relatively narrow region. Therefore, employing powerful attention mechanisms still delivers negligible guidance to the model, providing neither new structural details nor richer semantic cues—precisely the kind of insights our analytic perception and holistic perception are designed to offer.

4) Generality of Unc-SOD: By leveraging inherent uncertainty to mine qualified samples and mitigating hierarchical ambiguity via a novel perception-and-interaction pipeline, our Unc-SOD significantly boosts the performance of the pioneering two-stage detector, Faster R-CNN [10]. To demonstrate the generality of our approach, we integrate this uncertainty learning framework into two other representative proposal-based paradigms, with results presented in TABLE XVI.

Despite their formidable performance, our method still improves these strong baselines, Cascade R-CNN [11] and

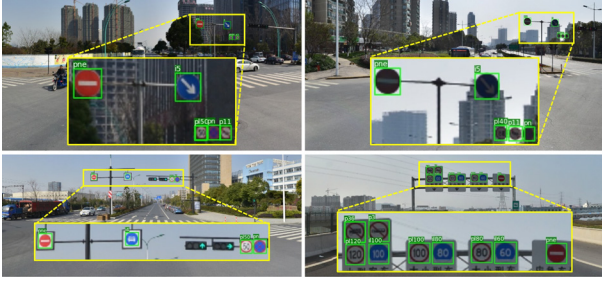


Fig. 12. Qualitative results of Unc-SOD on the TT100K test-set. Predictions with confidence scores larger than 0.3 are demonstrated. Best viewed in color and zoom-in windows.



Fig. 13. Failure cases of Unc-SOD, where it generates redundant predictions for small objects under heavy occlusion. Best viewed in dashed ellipses.

DetectoRS [88], by 1.4% and 1.5% points respectively, while also showing consistent gains on small-related metrics. This not only validates that the uncertainty we investigate is a prevalent issue within the two-stage detection paradigm, particularly for size-limited instances, but also underscores the generalizability of the framework to other two-stage approaches.

D. Qualitative Results

1) *Detection Visualization*: We showcase the qualitative results between Baseline [10] and our Unc-SOD on the SODA-D test-set in Fig. 11 as well as the detections on the TT100K test-set in Fig. 12. It can be seen that Unc-SOD handles small targets well, and it delivers precise bounding boxes even for these instances with extremely limited regions, while suppressing the false positives that plague the baseline detector. Moreover, the uncertainty distribution is derived from the entire dataset's training samples, which enables the model to dynamically allocate candidates to various instances while ensuring robust generalization across diverse scenarios.

2) *Failure Analysis*: To comprehensively understand our framework, we also visualize several failure results in Fig. 13. In extreme scenarios where small objects are heavily occluded, the close spatial proximity of ground-truth bounding boxes makes the instances highly indistinguishable. This severe spatial overlap hinders the potentially qualified priors, identified by our uncertainty-aware sampling strategy, from regressing precisely to their respective targets, thereby leading to redundant predictions that persist even after NMS.

V. CONCLUSION

In this paper, we present an Uncertainty learning framework for Small Object Detection, namely Unc-SOD. Firstly, we developed an Uncertainty-aware Sampling strategy to procure the indeterminacy for each instance which later on serves as a surrogate criterion to unearth potential samples for small objects, thereby alleviating the sample scarcity and preventing the optimization from being contaminated by the low-quality

priors of highly equivocal instances. Then, a Perception-and-Interaction pipeline is devised to leverage off-the-shelf cues to handle the feature inconsistency between two stages arising from the hierarchy-level uncertainty. Concretely, we constructed a mutual Perception scheme followed by the cross Interaction to exploit the intrinsic correlations in the region features from the original level and the assigned pyramid, thus compensating the limited information inherent in size-limited instances as well as deriving discriminative representations. Extensive experiments on the large-scale small object detection benchmarks SODA-D and SODA-A, along with several SOD-oriented datasets including COCO, Vis-Drone, and Tsinghua-Tencent 100K, underscore the superiority of our uncertainty-modeling paradigm for SOD task. Moreover, the proposed uncertainty learning framework currently focuses on the inherent data-level (aleatoric) uncertainty, while it is widely acknowledged that deep models can exhibit high model-level (epistemic) uncertainty when encountering challenging situations. Therefore, a promising future direction is to construct a more comprehensive, unified uncertainty learning pipeline, which could leverage aleatoric uncertainty to select high-quality samples, while using epistemic uncertainty to highlight valuable samples or deliver confidence calibration signal for the final detection, effectively suppressing overconfident false positives. This would contribute to a robust small object detection paradigm with a holistic understanding of uncertainty, and we leave this promising avenue for future research.

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